DATASHEET / LIST

Pipe Wall Thickness Cal & PMC Auto Generator

ASME B31.1 pipe thickness + multi CI-sheet generation. Outputs PMC results to Excel via allowable-stress table, material mapping, PDF parsing.

Key Purpose

Why this program

1. **Auto-generate ASME B31.1 thickness sheets**

   – Reads a CI-SHEET or P-T chart PDF, auto-detects the type, and computes required wall thickness per line

   → **Effect:** Hours of manual per-line thickness calculation collapse to one click, with no transcription error

2. **Produce the full Piping Material Class (PMC) book**

   – Fills a chosen PMC format with class data and outputs Excel (+PDF) ready for issue

   → **Effect:** A consistent, project-branded material spec is created in minutes instead of days of copy-paste

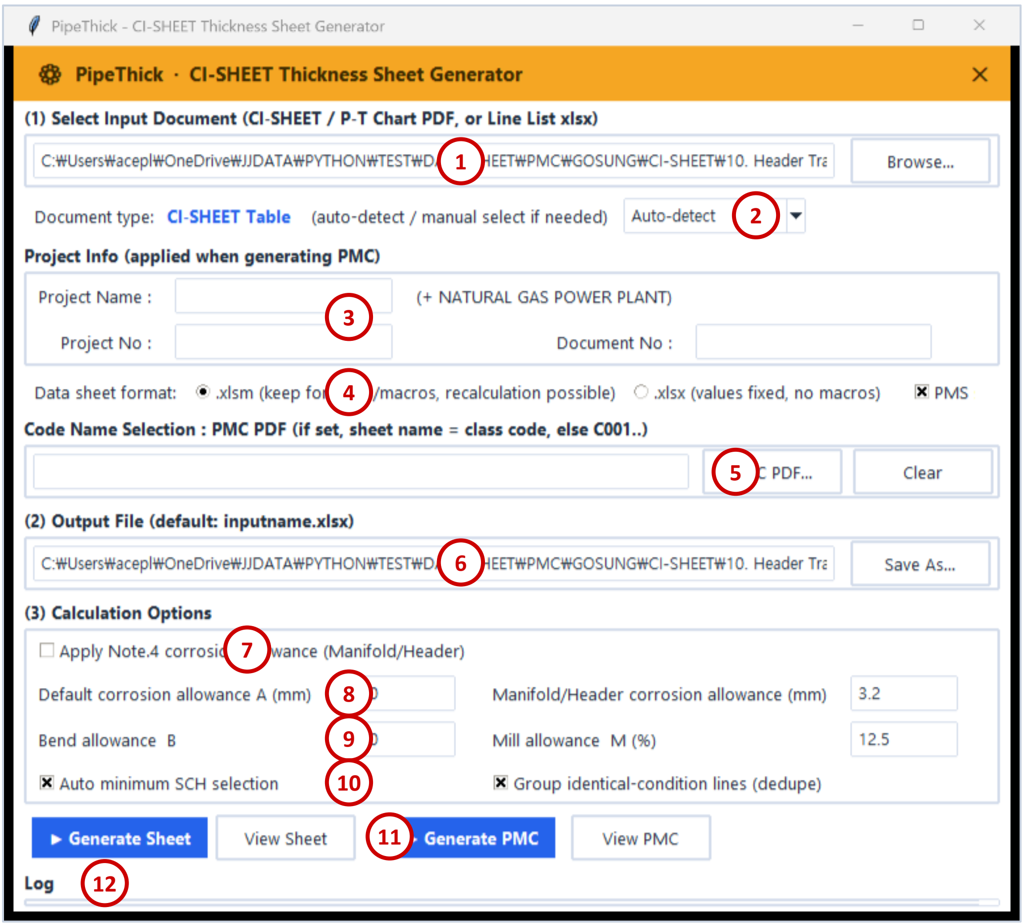
3. **Keep live formulas or cross-check a Line List**

   – .xlsm output stays recalculable; a Line List xlsx can be verified against the computed thickness

   → **Effect:** Downstream engineers can adjust inputs and re-verify without rebuilding the sheet

Required inputs : CI-SHEET / P-T Chart PDF (or Line List xlsx) + optional PMC PDF format + corrosion/bend/mill allowances

Main Window – Item Guide



1. Input document path (CI-SHEET / P-T Chart PDF or Line List) + Browse
2. Document type: auto-detect / manual select combobox
3. Project Info: Name / No / Document No (applied to PMC)
4. Data sheet format: .xlsm (formulas) vs .xlsx (fixed) + PMS sort
5. Code Name Selection: PMC PDF format entry + PMC PDF / Clear
6. Output file path (default inputname.xlsx) + Save As
7. Apply Note.4 Manifold/Header corrosion allowance toggle
8. Corrosion allowance A (mm) and Manifold/Header allowance (mm)
9. Bend allowance B and Mill allowance M (%)
10. Auto minimum SCH selection and Group identical lines (dedupe)
11. Generate Sheet / View Sheet / Generate PMC / View PMC
12. Log: progress and results

Overview

What it does

1. **ASME B31.1 pipe thickness**
2. **Multi CI-sheet generation**
3. **Outputs PMC results to Excel via allowable-stress table**
4. **Material mapping**
5. **PDF parsing**

Category: Datasheet / List
Folder: DATA_SHEET/PMC
Main scripts: pipethick_gui.py
Scripts / Status: 15 scripts · Operational
Tags: ASME B31.1 · PipeThick · Excel · PDF

Output Samples

Output Samples (1/4)

SECTION	LOCATION	QTY.	SIZE (IN)	MATERIAL	Design Temp. (℃)	Design Press. (bar.g) Note.2	MAWP (bar.g) Note.2	Remark
RH STEAM (Final-Stage Desuperheater)	HRH Main Steam Line Stop Check Valve ~ HRH Main Steam Line T.P.	1	24	SA335-P91	590	52.0	52.0	
	HRH Outlet Manifold ~ HRH Main Steam Line Stop Check Valve	1	24	SA335-P91	590	52.0	52.0	Inlcuding Stop Check Valve
HRH (SA213-T91)	HRH Outlet Manifold (Bottom)	1	24	SA335-P91	595	52.0	52.0	
	HRH Outlet Link	8	10	SA335-P91	595	52.0	52.0	
	HRH Outlet HDR (Bottom)	2	12	SA335-P91	595	52.0	52.0	
	HRH Inlet HDR (Top)	2	12	SA335-P91	575	52.0	52.0	
	HRH Inlet Link	8	10	SA335-P22	575	52.0	52.0	
	HRH Inlet Manifold (Top)	1	24	SA335-P22	575	52.0	52.0	
RH Inter-Stage Desuperheater	RHDS outlet	Same as HRH Inlet Manifold		SA335-P22	575	52.0	52.0	DRAIN POT CAP : 12" x MWT 36
	RH Inter-Stage Desuperheater	1	24	SA335-P22	575	52.0	52.0	
	RHDS Inlet	Same as CRH2 Outlet Manifold		SA335-P22	575	52.0	52.0	
CRH2 (SA213-T91)	CRH2 Outlet Manifold (Top)	1	24	SA335-P22	575	52.0	52.0	
	CRH2 Outlet Link	8	10	SA335-P22	575	52.0	52.0	
	CRH2 Outlet HDR (Top)	2	12	SA335-P91	575	52.0	52.0	
	CRH2 Inlet HDR (Bottom)	2	12	SA335-P91	534	52.0	52.0	
	CRH2 Inlet Link	8	10	SA335-P22	534	52.0	52.0	
CRH1 (SA213-T91)	CRH1 Outlet Link	Same as CRH2 Inlet Link		SA335-P22	534	52.0	52.0	
	CRH1 Outlet HDR (Bottom)	2	12	SA335-P91	534	52.0	52.0	
	CRH1 Inlet HDR (Top)	2	12	SA335-P22	444	52.0	52.0	
	CRH1 Inlet Link	8	10	SA335-P11	444	52.0	52.0	
	CRH1 Inlet Manifold (Top)	1	20	SA335-P11	444	52.0	52.0	
CRH INLET LINE (HP TBN EXHAUST)	CRH Steam Line 3 (IP steam Tie-in ~ CRH1 Inlet Manifold)	1	20	SA335-P11	444	52.0	52.0	
	CRH Steam Line 2 (CRH Steam Line MOV ~ IP steam Tie-in)	1	20	SA335-P11	444	52.0	52.0	
	CRH Steam Line 1 (HP TBN Exhaust ~ CRH Steam Line MOV)	1	20	SA335-P11	444	52.0	52.0	Including MOV
IP STEAM	IP Steam Line 3 (Check Valve ~ CRH Steam line Tie-in)	1	8	SA335-P11	444	52.0	52.0	Including Check Valve
	IP Steam Line 2 (PCV Downstream M/V ~ Check Valve)	1	8	SA106-B	338	52.0	52.0	
	IP Steam Line 1 (IPSH Outlet Manifold ~ PCV Downstream M/V)	1	8	SA106-B	338	53.0	53.0	Including M/V
IPSH (SA192)	IPSH Outlet Manifold (Bottom)	1	8	SA106-B	338	53.0	53.0	
	IPSH Outlet Link	4	4	SA106-B	338	53.0	53.0	
	IPSH Outlet HDR(Bottom)	2	6	SA106-B	338	53.0	53.0	
	IPSH Inlet HDR(Top)	2	6	SA106-B	273	53.0	53.0	
	IPSH Inlet Link (From Drum)	4	4	SA106-B	273	53.0	53.0	
IPEV (SA192) RISER	Header No.1 (High Gas temp.)	4	6	SA106-B	273	56.0	53.0	
	Header No.2	4	5	SA106-B	273	56.0	53.0	
IPEV (SA192) HDR	Top HDR (Out)	4	8	SA106-B	273	56.0	53.0	
					273	56.0	53.0	
	Bottom HDR (In)	4	8	SA106-B	273	56.0	53.0	
					273	56.0	53.0	
IPEV (SA192) FEEDER	Header No.1 (High Gas temp.)	6	4	SA106-B	273	56.0	53.0	2 EA는 Feeder Manifold Nozzle과 Direct connect
	Header No.2	6	3	SA106-B	273	56.0	53.0	2 EA는 Feeder Manifold Nozzle과 Direct connect
IPEV (SA192) FEEDER MANIFOLD	Feeder Manifold	2	10	SA106-B	273	56.0	53.0	
IPEV (SA192) DCMR	Downcomer	2	8	SA106-B	273	56.0	53.0	CCC CAP : 6" x Sch160
IEC2 TO DRUM	LCV downstream M/V ~ Drum Inlet	1	4	SA335-P11	328	66.3	53.0	
	LCV ~ LCV downstream M/V	1	4	SA335-P11	328	123.0	120.0	Including M/V
	FGH branch point ~ LCV	1	4	SA106-B	328	123.0	120.0	
	FGH branch point ~ FGH T.P	1	4	SA106-B	328	123.0	120.0	
	IEC2 Outlet Manifold ~ FGH branch point	1	4	SA106-B	328	123.0	120.0	
IEC2 (SA210-C)	IEC2 Outlet Manifold (Top)	1	4	SA106-B	328	123.0	120.0	
	IEC2 Outlet Link	4	3	SA106-B	328	123.0	120.0	
	IEC2 HDR 5"	2	5	SA106-B	328	123.0	120.0	
	IEC2 HDR 4"	2	4	SA106-B	328	123.0	120.0	
	IEC2 Inlet Link	4	3	SA106-B	328	123.0	120.0	
IEC1 (SA210-C)	IEC1 Outlet Link	Same as IEC2 Inlet Link		SA106-B	328	123.0	120.0	
	IEC1 HDR 5"	2	5	SA106-B	328	123.0	120.0	
	IEC1 HDR 4"	2	4	SA106-B	328	123.0	120.0	
	IEC1 Inlet Link	4	3	SA106-B	328	123.0	120.0	
	IEC1 Inlet Manifold (Top)	1	4	SA106-B	328	123.0	120.0	
IP FEEDWATER LINE	FW Line 3 (Check Valve ~ IEC1 Inlet Manifold)	1	4	SA106-B	328	123.0	120.0	Including Check Valve
	FW Line 2 (IP FW Branch point ~ Check Valve)	1	4	SA106-B	200	123.0	120.0	
	FW Line 1 (T.P ~ RH Inter/Final DSH Branch point)	1	4	SA106-B	200	123.0	120.0	
RH FINAL-STAGE DSH SPRAY WATER	Check valve ~ Spray nozzle	1	3	SA335-P91	590	68.0	52.0	Including Check Valve
	TCV downstream MOV ~ Check valve	1	3	SA106-B	200	68.0	52.0	
	RH Final DSH Branch point ~ TCV downstream M/V	1	3	SA106-B	200	123.0	120.0	Inlcuding M/V
RH INTER-STAGE DSH SPRAY WATER	Check valve ~ Spray nozzle	1	4	SA335-P22	575	68.0	52.0	Including Check Valve
	TCV downstream MOV ~ Check valve	1	4	SA106-B	200	68.0	52.0	
	RH Inter DSH Branch point ~ TCV downstream M/V	1	4	SA106-B	200	123.0	120.0	Including M/V

Note
1. Piping thickness shall be checked and input accordingly.
2. Design pressure for superheater = MAWP
   Design pressure for Evaporator/Economizer = MAWP + 3 bar(static head)
   Design pressure should be used for strength calculation.
3. Piping Corrosion Allowance is as follows ;
   - Alloy pipe : 1mm / C.S : 1.5mm
4. : HOLD
5. is revised.

CI-SHEET

8.0 CODE NAME.

C	A	1	B
(1)	(2)	(3)	(4)

(1) Class / Rating : See para. 8.1
(2) Material : See para. 8.2
(3) Sequence Number : See pare. 8.3
(4) Design Code : See pare. 8.4

8.1 Class / Rating

A : 4500# ASME B16.34
B : 2500# ASME B16.34 / B16.5
C : 1500# ASME B16.34 / B16.5
D : 900# ASME B16.34 / B16.5
E : 600# ASME B16.34 / B16.5
F : 400# ASME B16.5
G : 300# ASME B16.34 / B16.5
H : 150# ASME B16.34 / B16.5
X : Special as Designated

8.2 Material

A : Low, Intermediate And High Alloy Steel (Cr-Mo)
B : Carbon Steel (C.S)
C : Austenitic Stainless Steel (304 or 304L)
D : Austenitic Stainless Steel (316 or 316L)
E : Cast Iron or Ductile Iron(CI or DI)
F : Galvanized Carbon Steel
G : Copper
H : Polyvinyl Chloride(PVC) or Chlorinated Polyvinyl Chloride (CPVC)
J : Acrylonitrile Butadiene Styrene (ABS)
K : Plastic Lined Carbon Steel
L : High Density Polyethylene(HDPE)
M : Polypropylene(PP)
R : Fiberglass (Reinforced Plastic (FRP) - GRP, GRE, GRV, etc.)
S : Concrete

8.3 Sequence Number

This is a sequential number for the piping material classification.

8.4 Design Code

B : ASME B31.1 Boiler External Piping
(Use "SA" material based on ASME specification)
N : ASME B31.1 Non Boiler External Piping
("A" material based on ASTM or Equiv. specification can be used)

NAME CODE(PMC)

Output Samples

Output Samples (2/4)

No	Sheet	System / Location	NPS	DN	Material	Design Press(barg)	Design Temp(℃)	Req. Thk tr(mm)	Selected SCH
1	BA4	Check valve ~ Spray nozzle	4	DN 100	SA335-P22	68	575	11.21	Sch.160
2	BA1	HP Main Steam Line MOV ~ HP Main Steam Line T	12	DN 300	SA335-P92	189	589	37.29	A.W.T 38
3	BA2	HPSH4 Outlet Manifold (Top) (+1)	12	DN 300	SA335-P92	189	595	39.33	A.W.T 40
4	BA2_2	HPSH4 Outlet Link (+1)	6	DN 150	SA335-P92	189	595	20.44	XXS
5	BA2_3	HPSH4 Outlet HDR (Top) (+1)	8	DN 200	SA335-P92	189	595	26.61	A.W.T 27
6	DA2	HPSH3 Inlet HDR(Top)	8	DN 200	SA335-P91 (t ≤ 3")_(To'19	189	580	27.91	A.W.T 28
7	AA1	Check valve ~ Spray nozzle	3	DN 80	SA335-P92	240	589	12.66	XXS
8	AA1_2	Check valve ~ Spray nozzle	4	DN 100	SA335-P92	240	595	17.12	A.W.T 18
9	CB2	Riser	10	DN 250	SA106-C	192	367	22.13	Sch.140
10	CB2_2	Riser Manifold	14	DN 350	SA106-C	192	367	28.83	Sch.140
11	CB2_3	Header No.1 (High Gas temp.)	6	DN 150	SA106-C	192	367	13.64	Sch.120
12	CB2_4	Header No.2 (+4)	5	DN 125	SA106-C	192	367	11.45	
13	CB2_5	Top HDR (Out) (+1)	8	DN 200	SA106-C	192	367	17.76	Sch.120
14	CB2_6	Feeder Manifold	18	DN 450	SA106-C	192	367	37.05	Sch.140
15	CB2_7	Downcomer	16	DN 400	SA106-C	192	367	32.93	Sch.140
16	CB2_8	HEC5 Outlet Manifold ~ HP drum (+5)	12	DN 300	SA106-C	200	367	27.27	Sch.140
17	CB2_9	HEC5 Outlet Link (+7)	4	DN 100	SA106-C	200	367	9.63	Sch.120
18	CB2_10	HEC5 HDR 6" (+1)	6	DN 150	SA106-C	200	367	14.17	Sch.120
19	CB2_11	HEC5 HDR 5" (+1)	5	DN 125	SA106-C	200	367	11.9	
20	CB2_12	FW Line 4 (Check Valve ~ HEC1 Inlet Manifold) (+	10	DN 250	SA106-C	213	367	24.4	Sch.140
21	CA2	CRH2 Outlet Manifold (Top)	24	DN 600	SA335-P22	52	575	47.07	Sch.140
22	CA2_2	CRH2 Outlet Link	10	DN 250	SA335-P22	52	575	21.08	Sch.120
23	HB2	IP Steam Line 2 (PCV Downstream M/V ~ Check V	8	DN 200	SA106-B	52	444	9.64	Sch.60
24	HB2_2	IP Steam Line 1 (IPSH Outlet Manifold ~ PCV Dow	8	DN 200	SA106-B	53	444	9.82	Sch.60
25	DB1	FGH branch point ~ LCV (+8)	4	DN 100	SA106-B	123	328	6.54	Sch.80
26	DB2	FW Line 2 (IP FW Branch point ~ Check Valve) (+	4	DN 100	SA106-B	123	200	6.54	Sch.80
27	DA4	LCV ~ LCV downstream M/V	4	DN 100	SA335-P11	123	328	7.16	Sch.80
28	DA3	CRH2 Inlet Link	10	DN 250	SA335-P22	52	534	13.24	Sch.80
29	DA1	HRH Main Steam Line Stop Check Valve ~ HRH M	24	DN 600	SA335-P91 (t ≤ 3")_(To'19	52	590	27.23	Sch.80
30	DA1_2	HRH Outlet Manifold (Bottom)	24	DN 600	SA335-P91 (t ≤ 3")_(To'19	52	595	29.04	Sch.80
31	DA1_3	HRH Outlet Link	10	DN 250	SA335-P91 (t ≤ 3")_(To'19	52	595	13.01	Sch.80
32	DA1_4	HRH Outlet HDR (Bottom)	12	DN 300	SA335-P91 (t ≤ 3")_(To'19	52	595	15.43	Sch.80
33	DA1_5	CRH2 Outlet HDR (Top)	12	DN 300	SA335-P91 (t ≤ 3")_(To'19	52	575	12.1	XS
34	DA2_2	Check valve ~ Spray nozzle	3	DN 80	SA335-P91 (t ≤ 3")_(To'19	68	590	5.11	Sch.40
35	EB2	IPSH Outlet Manifold (Bottom)	8	DN 200	SA106-B	53	338	5.53	Sch.20
36	EB2_2	IPSH Outlet Link	4	DN 100	SA106-B	53	338	2.88	Sch.40
37	EB2_3	IPSH Outlet HDR(Bottom)	6	DN 150	SA106-B	53	338	4.25	Sch.40
38	EB2_4	IPSH Inlet HDR(Top)	6	DN 150	SA106-B	53	273	4.25	Sch.40
39	EB2_5	IPSH Inlet Link (From Drum)	4	DN 100	SA106-B	53	273	2.88	Sch.40
40	EB1	Header No.1 (High Gas temp.)	6	DN 150	SA106-B	56	273	4.48	Sch.40
41	EB1_2	Header No.2	5	DN 125	SA106-B	56	273	3.76	
42	EB1_3	Top HDR (Out) (+2)	8	DN 200	SA106-B	56	273	5.84	Sch.20
43	EB1_4	Header No.1 (High Gas temp.)	4	DN 100	SA106-B	56	273	3.04	Sch.40
44	EB1_5	Header No.2	3	DN 80	SA106-B	56	273	2.37	Sch.40
45	EB1_6	Feeder Manifold	10	DN 250	SA106-B	56	273	7.27	Sch.30
46	EB3	CPH Bypass Line (+1)	6	DN 150	SA106-B	43	257	3.46	Sch.40
47	EB3_2	TCV Downstream M/V ~ CPH Inlet Tie-in (+2)	8	DN 200	SA106-B	43	257	4.5	Sch.20

Auto Pipe Thk Cal Index

PIPE WALL THICK. CA << INDEX

Rev.	0
Sheet	1 of 1

SYSTEM : Check valve ~ Spray nozzle   PMC :

Design Value

Design Press. P	68.0	barg
Design Temp. T	575.0	℃

Selected Material

Code	**ASME SECTION Ⅱ ('23)**
Material Group	Low and Intermediate Alloy Steel
Specification	**SA335-P22**

Description	Value			Source or Remark
Selected Pipe	NPS=	4	IN.	DN 100
Outside Diameter of Pipe	Do =	114.30	mm	
Allowable Stress Value	SE =	5,054	psi	at Design Temperature 1067 °F
Values of y	y =	0.70		Table 104.1.2(A)
Additional Thickness	A =	-	mm	Clause 104.1.2
Bend Thining Allowance	B =	1.00		Table 102.4.5 or Manufacturer's recommendation
Min. Wall Thk	tm=	9.81	mm	Clause 104.1.2 tm = (P * Do) / (2 (SE + Py)) + A
Min. Wall Thk with Bending	tm=	9.81	mm	Table 102.4.5 or Manufacturer
Mill Allowance	M =	12.50	%	
Required Nominal Wall Thk	tr=	11.21	mm	
Valve Rating	**CL.2500**			Pressure Margin 19.7 barg

Pipe Thickness Selection Tables for various NPS

Selected Pipe				Required Thickness						Remark	
NPS inch	Do mm	SCH.	tn	y	A	B	M	tm	tr	(tr/tn)	y'
1/2	21.34	Sch.40	**2.77**	0.7	0	1	12.5	1.83	**2.09**	0.76	0 OK
3/4	26.67	Sch.40	**2.87**	0.7	0	1	12.5	2.29	**2.62**	**0.91**	0 OK
1	33.40	Sch.40	**3.38**	0.7	0	1	12.5	2.87	**3.28**	**0.97**	0 OK
1 1/2	48.26	Sch.80	**5.08**	0.7	0	1	12.5	4.14	**4.73**	**0.93**	0 OK
2	60.33	Sch.160	**8.71**	0.7	0	1	12.5	5.18	**5.92**	0.68	0 OK
2 1/2	73.03	Sch.160	**9.52**	0.7	0	1	12.5	6.27	**7.16**	0.75	0 OK
3	88.90	Sch.160	**11.13**	0.7	0	1	12.5	7.63	**8.72**	0.78	0 OK
4	114.30	Sch.160	**13.49**	0.7	0	1	12.5	9.81	**11.21**	0.83	0 OK
6	168.30	Sch.160	**18.24**	0.7	0	1	12.5	14.45	**16.51**	**0.91**	0 OK
8	219.10	XXS	**22.22**	0.7	0	1	12.5	18.81	**21.50**	**0.97**	0 OK
10	273.05	Sch.160	**28.57**	0.7	0	1	12.5	23.44	**26.79**	**0.94**	0 OK
12	323.85	Sch.160	**33.32**	0.7	0	1	12.5	27.80	**31.77**	**0.95**	0 OK
14	355.80	Sch.160	**35.71**	0.7	0	1	12.5	30.54	**34.91**	**0.98**	0 OK
16	406.40	Sch.160	**40.46**	0.7	0	1	12.5	34.89	**39.87**	**0.99**	0 OK

관련 Table On/Off

ASME 연도별 허용 응력표

DIN 자재 허용 응력표

16.34 Valve Rating 표

Valve Rating Selection

Mat. Group 1.10
A182-F22, A217-WC9

Valve 재질은 Pipe재질과
연동되지 않으므로
필히 확인 바랍니다.

A105 및
Sutainless(304,316등)
자재에 대한 Rating은

Detail data for Valve rating

Valve Rating		CL.2500	
Low Temp	1,050.0	1,455.0	psig
Dgn Temp	**1,067.0**	**1,271.4**	psig
High Temp	1,100.0	915.0	psig
Pressure Margin		**285.1**	psig
		20.0	kg/㎠g
		19.7	barg

Design Pressure / Temp.

Dgn P	986.3 psig
Dgn T	1,067.0 °F

Allowable Stress Value according to Code

	Temp.(°F)		
	1,050.0	1,067.0	1,100.0
SE(psig)	5,700.0	5,054.0	3,800.0

Time Dependent

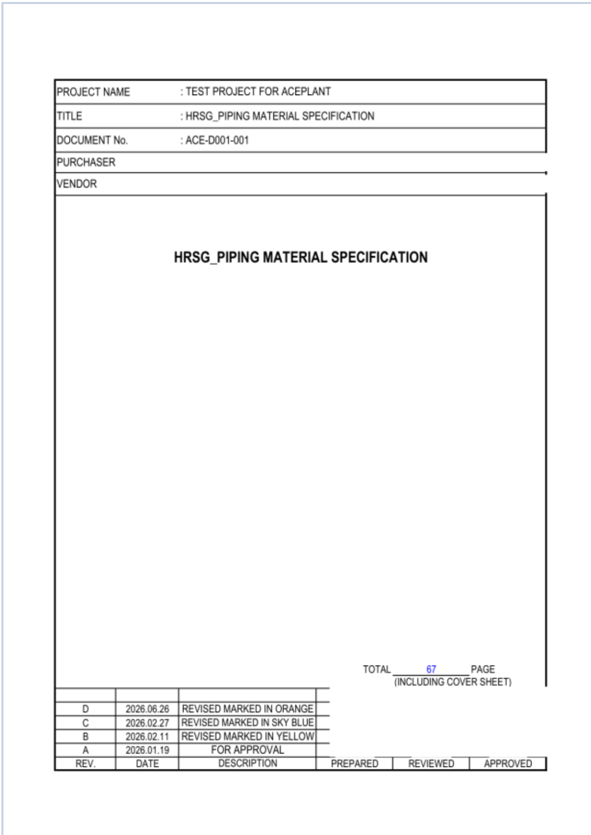
ASME SECTION Ⅱ ('23) SA335-P22   T4

TABLE 2 CORRELATION OF FITTINGS CLASS WITH SCHEDULE NUMBER OR WALL DESIGNATION OF PIPE FOR CALCULATION OF RATINGS

Pipe Wall Thick Calculation Sheet

Output Samples

Output Samples (3/4)



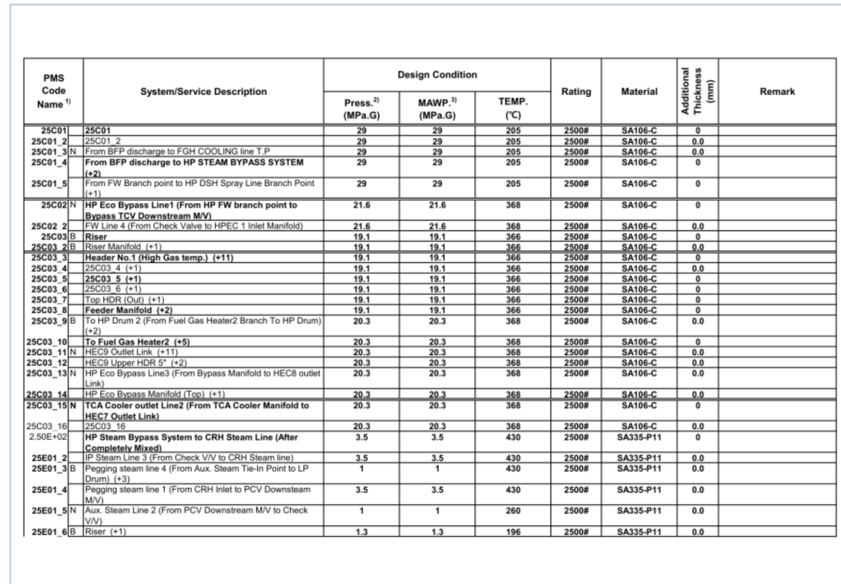
PROJECT NAME	: TEST PROJECT FOR ACEPLANT
TITLE	: HRSG_PIPING MATERIAL SPECIFICATION
DOCUMENT No.	: ACE-D001-001
PURCHASER	
VENDOR	

HRSG_PIPING MATERIAL SPECIFICATION

TOTAL 67 PAGE
(INCLUDING COVER SHEET)

REV.	DATE	DESCRIPTION	PREPARED	REVIEWED	APPROVED
D	2026.06.26	REVISED MARKED IN ORANGE			
C	2026.02.27	REVISED MARKED IN SKY BLUE			
B	2026.02.11	REVISED MARKED IN YELLOW			
A	2026.01.19	FOR APPROVAL			

PMC Auto Generator Sample 1



PMS Code Name [1]		System/Service Description	Design Condition Press. [2] (MPa.G)	MAWP. [3] (MPa.G)	TEMP. (℃)	Rating	Material	Additional Thickness (mm)	Remark
25C01		**25C01**	29	29	205	2500#	SA106-C	0	
25C01_2		25C01_2	29	29	205	2500#	SA106-C	0.0	
25C01_3	N	From BFP discharge to FGH COOLING line T.P	29	29	205	2500#	SA106-C	0.0	
25C01_4		**From BFP discharge to HP STEAM BYPASS SYSTEM (+2)**	29	29	205	2500#	SA106-C	0	
25C01_5		From FW Branch point to HP DSH Spray Line Branch Point (+1)	29	29	205	2500#	SA106-C	0	
25C02	N	**HP Eco Bypass Line1 (From HP FW branch point to Bypass TCV Downstream M/V)**	21.6	21.6	368	2500#	SA106-C	0	
25C02_2		FW Line 4 (From Check Valve to HPEC 1 Inlet Manifold)	21.6	21.6	368	2500#	SA106-C	0.0	
25C03	B	**Riser**	19.1	19.1	366	2500#	SA106-C	0	
25C03_2	B	Riser Manifold (+1)	19.1	19.1	366	2500#	SA106-C	0.0	
25C03_3		**Header No.1 (High Gas temp.) (+11)**	19.1	19.1	366	2500#	SA106-C	0	
25C03_4		25C03_4 (+1)	19.1	19.1	366	2500#	SA106-C	0.0	
25C03_5		**25C03_5 (+1)**	19.1	19.1	366	2500#	SA106-C	0	
25C03_6		25C03_6 (+1)	19.1	19.1	366	2500#	SA106-C	0	
25C03_7		Top HDR (Out) (+1)	19.1	19.1	366	2500#	SA106-C	0	
25C03_8		**Feeder Manifold (+2)**	19.1	19.1	366	2500#	SA106-C	0	
25C03_9	B	To HP Drum 2 (From Fuel Gas Heater2 Branch To HP Drum) (+2)	20.3	20.3	368	2500#	SA106-C	0.0	
25C03_10		**To Fuel Gas Heater2 (+5)**	20.3	20.3	368	2500#	SA106-C	0	
25C03_11	N	HEC9 Outlet Link (+11)	20.3	20.3	368	2500#	SA106-C	0.0	
25C03_12		HEC9 Upper HDR 5" (+2)	20.3	20.3	368	2500#	SA106-C	0.0	
25C03_13	N	HP Eco Bypass Line3 (From Bypass Manifold to HEC8 outlet Link)	20.3	20.3	368	2500#	SA106-C	0.0	
25C03_14		HP Eco Bypass Manifold (Top) (+1)	20.3	20.3	368	2500#	SA106-C	0.0	
25C03_15	**N**	**TCA Cooler outlet Line2 (From TCA Cooler Manifold to HEC7 Outlet Link)**	20.3	20.3	368	2500#	SA106-C	0	
25C03_16		25C03_16	20.3	20.3	368	2500#	SA106-C	0.0	
2.50E+02		**HP Steam Bypass System to CRH Steam Line (After Completely Mixed)**	3.5	3.5	430	2500#	SA335-P11	0	
25E01_2		IP Steam Line 3 (From Check V/V to CRH Steam line)	3.5	3.5	430	2500#	SA335-P11	0.0	
25E01_3	B	Pegging steam line 4 (From Aux. Steam Tie-In Point to LP Drum) (+3)	1	1	430	2500#	SA335-P11	0.0	
25E01_4		Pegging steam line 1 (From CRH Inlet to PCV Downsteam M/V)	3.5	3.5	430	2500#	SA335-P11	0.0	
25E01_5	N	Aux. Steam Line 2 (From PCV Downstream M/V to Check V/V)	1	1	260	2500#	SA335-P11	0.0	
25E01_6	B	Riser (+1)	1.3	1.3	196	2500#	SA335-P11	0.0	

PMC Auto Generator Sample 2

Output Samples

Output Samples (4/4)

CLASS NO.	25C01		
RATING	2500#	MATERIAL	CARBON STEEL

DESIGN CONDITION PRESS. (MPa.G)	MAWP (MPa.G)	TEMP. (℃)	Code Suffix	SYSTEM/SERVICE DESCRIPTION	DESIGN CONDITION PRESS. (MPa.G)	MAWP (MPa.G)	TEMP. (℃)	Code Suffix	SYSTEM/SERVICE DESCRIPTION
29	29	205		**25C01**					
			0	25C01_2					
			N	From BFP discharge to FGH COOLING line T.P					
29	29	205		**From BFP discharge to HP STEAM BYPASS SYSTEM (+2)**					
			0	From FW Branch point to HP DSH Spray Line Branch Point (+1)					

PIPE AND FITTING

ITEM	SIZE (Inch)	SCH. NO. / RATING	MATERIAL	CONN'	DIMENSIONS
PIPE	1/2 ~ 2	Sch.160	SA106-C	PE, SMLS	ASME B36.10M
	2 1/2 ~ 6	XXS	SA106-C	BE, SMLS	ASME B36.10M
	8	A.W.T 25	SA106-C	BE, SMLS	ASME B36.10M
	10	A.W.T 31	SA106-C	BE, SMLS	ASME B36.10M
	12	A.W.T 36	SA106-C	BE, SMLS	ASME B36.10M
	14	A.W.T 40	SA106-C	BE, SMLS	ASME B36.10M
FITTING [2]	1/2 ~ 2	6000#	SA105	SW	ASME B16.11
	2 1/2 ~ 6	XXS	SA234-WPC	BW, SMLS	ASME B16.9
	8	A.W.T 25	SA234-WPC	BW, SMLS	ASME B16.9
	10	A.W.T 31	SA234-WPC	BW, SMLS	ASME B16.9
	12	A.W.T 36	SA234-WPC	BW, SMLS	ASME B16.9
	14	A.W.T 40	SA234-WPC	BW, SMLS	ASME B16.9
FLANGE [4]	1/2 ~ 2	2500#	SA105	SW, RF	ASME B16.5
	2 1/2 ~ 12	2500#	SA105	WN, RF	ASME B16.5

VALVE [1]

SIZE (Inch)	MATERIAL BODY	MATERIAL STEM / TRIM	RATING	CONN'
GATE / GLOBE / CHECK				
1/2 ~ 2	SA105	13Cr or Equiv. / STL	2500#	SW
2 1/2 ~ 14	SA216-WCC OR SA105	13Cr or Equiv. / STL	2500#	BW
(BLANK)				
(BLANK)				

MISCELLANEOUS

GASKET	ALL	2500#	SPIRAL WOUND, 4.5T, 316SS WITH GRAPHITE HOOP&FILLER, 316SS IN/OUT RING
BOLT	ALL	STUD BOLT : SA193-B16	
NUTS	ALL	HEAVY HEX. NUTS : SA194-3	

NOTE

1) Refer to the valve list for details.
2) Fitting Thickness shall be equal to or greater than pipe.
4) SRV in/out flange rating shall be applied in accordance with the vender selected specification.
(In case WCC is applied by SRV, the use of A105 flange by piping shall be re-checked if there need to be upgraded to F11 or not.)

PMC Auto Generator Sample 3

Notes & Information

- Part of the in-house Plant Engineering automation suite — category: Datasheet / List.
- Runs offline as a pure-Python / Tkinter tool; portable across PCs via OneDrive sync (needs Python + packages).
- Launch from the PC launcher (런처.bat) or run the main script: pipethick_gui.py.
- Output samples and the program window above reflect the current build.

Thank You

Pipe Wall Thickness Cal & PMC Auto Generator · Plant Engineering Total Service System